INSTITUT FOR MATEMATIK
AARHUS UNIVERSITET

Ute Hahn
November 6, 2026

Statistical Models

Week 11: Metropols-Hastings algorithm, rjags, diagnosing convergence

> ! Extended lectures on Wednesday
>
> Wednesday in this week 11 (November 12th), we will again have longer of lectures, that is from 14-17.
> This is to make up for the fact that there were no lectures on Wednesday, November 5th (course week 10).

In the Wednesday lecture, we introduce the *Metropolis-Hastings-algorithm* ("MH"). We start with the general principle and an example for its implementation. The algorithm has an interesting history; if time allows, we will also touch upon that. Then we discuss variants of MH-sampling: we will see that using a symmetric proposal density makes calculations a bit simpler. One can also combine Gibbs sampling with MH, which is important in Bayesian analysis for multivariate parameters, when the priors are not conjugate to the likelihood.

All these algorithms can be used "out of the box' '; there are various software options. You will learn to use the software JAGS for Bayesian data analysis. To this end, you need to describe your model in a well defined way; I will discuss the general structure of JAGS model descriptions and we look at a few examples. You can download the JAGS user manual here and on brightspace.

If time allows, we look at statistics that help to assess if the Markov chains have converged and are mixing well: *trace plots*, the *Geweke* and *Gelman statistics*. The *effective sample size* tells us how much information we get out of a long Markov chain as compared to independent sampling. This is usually less, due to autocorrelation. We will discuss points to consider if we want to improve convergence speed and decrease autocorrelation (R&G, Section 3.4.2, 3.4.3).

Relevant Readings

- ☐ Metropolis-Hastings algorithm: R&G, Section 3.2.2,
- ☐ JAGS: R&G, Section 3.3
- ☐ Assessing and improving convergence speed: R&G, Section 3.4, in particular 3.4.2 and 3.4.3

Install JAGS

From this week on, you will use the R package `rjags` for Bayesian analysis. `rjags` is an interface to the software JAGS (Just Another Gibbs Sampler) that you need to install separately on your computer (not from within R). Please install JAGS from sourceforge and feel free to ask for help if you experience problems with that.

Lecture Material

- Link to onenotes on Google drive
- Coding the Metropolis Hastings algorithm

Exercises 11

Problem 11.1: A model for new species discoveries (Metropolis-Hastings sampling)

As discussed in Reich & Ghosh, Section 1.6, authors Edie, Smits and Jablonski (2017, reference 24 in R&G) report that the number of marine bivalve species discovered each year from 2010-2015 was $64, 13, 33, 18, 30$ and 20.
Denote Y_t as the number of species discovered in year $2009 + t$ (so that $Y_1 = 64$ is the count for 2010).
Consider the following model for these data:

$$Y_t \mid \alpha, \beta \overset{\text{indep}}{\sim} \text{Poisson}(\lambda_t),$$

where $\lambda_t = \exp(\alpha + \beta t)$ and $\alpha, \beta \overset{\text{indep}}{\sim} \text{Normal}(0, 10^2)$.

a. Use JAGS to fit the model. Summarize the posterior of α and β and verify that the MCMC sampler has converged. Does this analysis provide evidence that the rate of discovery is changing over time?

b. Write your own Metropolis sampler (i.e., not JAGS). Comment your code, report your candidate distribution for each parameters and corresponding acceptance ratio, and use trace plots for each parameter to show the chains have converged.

see Reich & Ghosh, Chapter 3, Exercises 10 and 11.

Problem 11.2: Bayesian analysis of galaxies

see Reich & Ghosh, Chapter 3, Exercise 7.
Open and plot the galaxies data in R using the code below,

```
library(MASS)
data(galaxies)
?galaxies
Y <- galaxies
n <- length(Y)
hist(Y, breaks=25)
```

Model the observations $Y_1, \ldots, Y_{82}$ using the Student-t distribution with location μ, scale σ and degrees of freedom k.
? What is this distribution? Here we mean that $Y = \sigma \cdot Z + \mu$, where Z has a standard Student-t distribution with k degrees of freedom.
Assume prior distributions $\mu \sim \text{Normal}(0, 10000^2)$, $1/\sigma^2 = \tau \sim \text{Gamma}(0.01, 0.01)$ and $k \sim \text{Uniform}(1, 30)$.

a. Give reasonable initial values for each of the three parameters.
b. Fit the model using JAGS. Report trace plots of each parameter and discuss convergence.
c. Graphically compare the t distribution with parameters set to their posterior mean with the observed data. Does the model fit the data well?

Problem 11.3: MCMC for Bayesian analysis of shifted exponential variables

see Reich & Ghosh, Chapter 3, Exercise 14.

Let $Y_1, \ldots, Y_n$ be a random sample from a shifted exponential distribution with density

$$f(y \mid \alpha, \beta) = \begin{cases} \beta \exp(-\beta(y-\alpha)), & y \geq \alpha \\ 0 & y < \alpha \end{cases}$$

where $\alpha > 0$ and $\beta > 0$ are the parameters of interest. Assume prior distributions $\alpha \sim \text{Uniform}(0, c)$ and $\beta \sim \text{Gamma}(a, b)$.

a. Plot $f(y \mid \alpha, \beta)$ for $\alpha = 2$ and $\beta = 3$ and $y \in [0, 5]$, give interpretations of α and β, and describe a real experiment where this model might be appropriate.

b. Give the full conditional distributions of α and β and determine if they are members of a common family of distributions.

c. Write pseudo code for an MCMC sampler including initial values, full details of each sampling step, and how you plan to assess convergence.

If you have time, also implement your MCMC sampler, and check if it gives reasonable results.

Problem 11.4: Short discussion and comparison of numerical methods

Give an advantage and a disadvantage of the following methods:

a. Maximum a posteriori estimation
b. Numerical integration
c. Bayesian central limit theorem
d. Gibbs sampling
e. Metropolis-Hastings sampling

Reich & Ghosh, Chapter 3, Exercise 1.

Problem 11.5: Compare your own Gibbs sampler results to JAGS output

In Exercises 10.3 and 10.4, you were asked to code a Gibbs sampler "by hand". If you discussed (at least one of) these exercises in class, try to redo the calculations with JAGS, and compare your results with the output you get.
Reich & Ghosh, Chapter 3, Exercises 4(e) and/or 5(d)

Problems 11.1, 11.2 and 11.5 include hands on calculations with JAGS. Feel encouraged to try at least one of them :-)